

Roll No.

TCE-801

B. TECH. (CE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023
EARTHQUAKE RESISTANT DESIGN OF BUILDINGS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

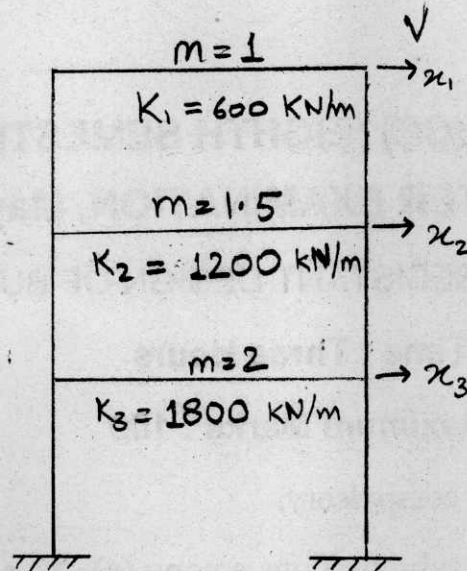
(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Find the original mass and stiffness of undamped SDOF system which is subjected to harmonic excitation where resonance occurs when exciting frequency is 6 rad/s. When the mass is increased by 1 kg resonance is found to be occurring at 5.86 rad/s. (CO1)
- (b) Find the equation of motion for a SDOF system which is free and undamped. (CO1)
- (c) A mass (m) is placed at one end of cantilever beam of length (l) and flexural rigidity (El) having negligible mass. Determine its natural frequency. (CO1)

P. T. O.

2. (a) Define natural frequency, normal mode of vibration, fundamental frequency, fundamental mode of vibration and resonance. (CO2)
- (b) Write the equation of motion in matrix form for given system : (CO2)



- (c) How can we transform differential equations of MDOF system into independent equation having only one dependent variable using modal superposition method ? (CO2)
3. (a) Differentiate between primary and secondary seismic waves with the help of diagrams. (CO3)
- (b) Explain plate tectonic theory responsible for the formation of seismic waves. (CO3)
- (c) Mention the factors influencing the intensity of earthquake at a particular place. Define the intensity of earthquake and explain how the intensity can be measured in Modified Mercalli Intensity Scale. (CO4)

(3)

4. (a) Discuss the role of ductility in earthquake resistant design of buildings and how it can be improved in individual structural members as well as whole structure. (CO4)
- (b) Mention the role of response spectrum in designing earthquake resistant RC buildings. (CO4)
- (c) Differentiate between response spectrum and design spectrum. Mention the need of smoothened elastic design response spectrum. (CO4)
5. (a) Discuss the mitigating effect of earthquakes on structures. (CO5)
- (b) Explain the process of determining the design base shear for a building using equivalent static lateral load method IS 1893 (part 1) : 2002. (CO5)
- (c) Elaborate the purpose, principles and advantages of base isolation. (CO5)

Roll No.

TCE-813

**B. TECH. (CIVIL ENGINEERING) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

HYDROPOWER ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the strength and opportunity of a waterpower in India. (CO1)

(b) Predict the load of a town in year 2021 using Scheer's formula and the following data : (CO1)

Per capita load in year 2018 = 450 kWh

Population growth rate = 8% (constant for five years period)

Population in year 2018 = 60,00,000

(c) A four way is to design to meet out the pondage requirement of a runoff river scheme. The daily flow in the river is $20 \text{ m}^3/\text{s}$. The plant is to be used as 15 hrs. peaking station.

Calculate the pondage factor and storage capacity of four ways. (CO1)

P. T. O.

2. (a) Calculate the specific speed of a turbine and suggest the type of turbine required for a river having a discharge of 240 L/s with an available head of 45 m. Take the efficiency of turbine = 82% and speed = 450 r.p.m.

(CO2)

- (b) Explain the Diversion Canal Plant with different layouts with a neat sketch.

(CO2)

- (c) Run-of-river plant on a stream has inflow of 20 cumecs and net head of 30 m with provision for pondage to meet daily peak demand with a load factor of 60%. Calculate :

(CO2)

- (i) The power generation capacity of the plant at 80% overall efficiency.

- (ii) The plant runs as peaking station for 3 hrs. and balance period in the day for average load.

Determine the amount of pondage needed.

3. (a) Describe the *three* main divisions of a power station. Explain in detail with a neat sketch.

(CO3)

- (b) Explain underground power station? Describe the history of underground power station.

(CO3)

- (c) Explain all the types of an underground power station with a neat sketch.

(CO3)

4. (a) Describe the 'economical diameter of a penstock'. Also explain the different methods of calculating the economical diameter of a penstock.

(CO4)

- (b) A hydroelectric station is not to be supplied with $10 \text{ m}^3/\text{sec}$ of water through a penstock which has a friction factor of 0.016. The maximum normal head of the penstock is 50 kg/cm^2 and a water hammer overpressure of 20% over the normal pressure is anticipated. The safe head in the steel used is presumed to be 3000 kg/cm^2 . The ready penstocks at site are likely to cost ₹ 15,000 per tonne including erection charges. The life of the project is 50 years and the rate of interest is 7 per cent. It is proposed to sell the energy at the rate of Re 0.03/kWh. Predict the optimum diameter of the penstock, given the OMR costs are 5 per cent. Assume the turbine efficiency to 90 per cent and the annual load factor is 0.4. (CO4)
- (c) The Tallulah Falls Project in USA has a surge tank at the end a 2030 m long tunnel, 4.22 m in dia. The surge tank, which is rectangular in section, may be presumed to be of a circular cross-section, 15.85 m in diameter. The penstock system consists of 5 penstock each 1.525 m in dia. and 380 m long. These can be represented by a single penstock, 380 m long and 3.41 m in diameter. Friction factor f for tunnel and the penstock in 0.018 and 0.03 respectively. The acoustic wave velocity in the penstock is 1370 m/s. In steady state, the head reservoir level is E 1.457.00 with a discharge of $26.2 \text{ m}^3/\text{s}$. Determine the water hammer pressure for a sudden closure. (CO4)
5. (a) Explain a draft tube. Draft tube applicable in the installation of turbines. Explain in detail. (CO5)

(b) Explain governing of turbine. Also describe the hydraulic governing.

(CO5)

(c) At the Volgaimeni V. I. Lenin Hydroelectric Plant, in USSR, the Kaplan turbine used has the following data :

Operation head is 22.5 m, Output power at this head is 126 MW, Discharge at this head is $615 \text{ m}^3/\text{s}$, Speed is 68.2 r.p.m., Runner tip-to-tip diameter (D) is 9.3 m, Hub diameter (D_H) is 4.3 m, Number of blades is 6.

Calculate the speed ratio, the flow ratio and the overall efficiency and the maximum suction draft head.

(CO5)

Roll No.

TCE-851

B. TECH. (CE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023

DISASTER MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Classify natural disasters that your country deals with and enlist various such disasters that your region (State) has faced in past as per their year of occurrence. (CO1)
- (b) How would you describe "Hazards" and "Vulnerability" ? Write *five* possible causes of forest fires. (CO1)
- (c) Rephrase the meaning of "Mitigation" and being a civil engineer how your discipline could help in mitigating a disaster. (CO1)
2. (a) Can you make use of the fact that the man-made disasters have a long-term impact than a natural disaster on the environment ? Explain. (CO2)

P. T. O.

(2)

- (b) Explain the term Sustainable Development goals. How can they be achieved ? (CO2)
- (c) What elements would you choose to explain differential impacts of a disaster ? (CO2)
- 3. (a) What facts would you suggest explaining positive realms of a disaster on the development ? (CO3)
- (b) What elements can you list to explain prediction, forecasting and warning in context to disaster management ? (CO3)
- (c) How can climate change be mitigated ? (CO3)
- 4. (a) Are you aware with the functions of the rescue party and importance of various types of rescue training ? (CO4)
- (b) "Land zoning can reduce the disaster." Justify the statement in your words. (CO4)
- (c) Under post-disaster management, kindly highlight the urban risks and urban disaster mitigation strategies in detail. (CO4)
- 5. (a) Necessity of DMA. List down the government bodies that are associated with DMA. (CO5)
- (b) Consider a case study of forest fire of Uttarakhand and being the head of NEC what will be the flow chart that you will follow to mitigate this disaster ? (CO5)
- (c) Consider a case study of flash floods in Uttarakhand and being the head of NEC, what will be the flow chart that you will follow to mitigate this disaster ? (CO5)